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RESEARCH ARTICLE

Application of cloud computing and big data in three-stage dynamic modeling of disaster relief logistics and wounded transportation: a case study

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Abstract
Collecting and sharing information about affected areas is an important activity for optimal decision-making in relief processes. Defects such as over-sending some items to affected areas and mistakes in transferring injured people to medical centers in accidents are due to improper management of this information. Because cloud computing as a processing and storage platform for big data is independent of the device and location and can also perform high-speed processing, its use in disasters has been highly regarded by researchers. In this environment, a three-stage dynamic procedure for evacuation operations and logistics issues is presented. The first stage of the proposed model is image processing and tweet mining in a cloud center in order to determine the disaster parameters. In stage II, a mixed-integer multi-commodity model is presented for the relief commodity delivery, wounded people transportation with capacity constraints, and locating of the possible on-site clinics and local distribution centers near disaster areas. In stage III, by using a system of equations, detailed vehicle load/unload instructions are obtained. Finally, the effectiveness of the proposed model on the data of an earthquake disaster in Iran is investigated. The results of comparing the proposed approach with a two-stage algorithm show that the total number of unsatisfied demand for all types of commodities in the proposed approach was better than the other. Also, the number of survivors in the three-stage model is significantly higher than in the two-stage one. The better performance of the proposed algorithm is due to the fact that online data is continuously available and that decisions such as sending relief items and dispatching are made more effectively.

Keywords Disaster logistics · Evacuation · Cloud computing · Relief distribution · Multi-stage modeling

Introduction

A disaster is a sudden and catastrophic event that causes human, economic, or environmental damage. Between 1980 and 2020, approximately 7 billion people have been affected by natural disasters, resulting in a reported loss of 3400 billion dollars (EM-DAT 2020). Also, natural disasters, in addition to human and economic losses, lead to psychological damage to survivors (Zhang et al. 2022). Disaster risk reduction is an integral part of social and economic development in the mainstream of sustainable development, and this issue has been recognized by several international documents such as Sustainable Development Goals (SDGs). For example, the goals on resilience and sustainability in cities under SDG#11 and combating climate change and its impacts under SDG#13 reaffirm the interrelationship between disaster risk reduction and sustainable development. In logistical support and evacuation of the injured people at the time of the disaster, quick response to urgent needs of relief, and transfer of injured people are essential to reduce the effects of the disaster (Thakur et al. 2022). In this regard, the evacuation of wounded people happens within the early response phase, while logistics assistance activities are continued for responding to the needs of people in the affected regions. Another important issue is the information about disaster areas that can improve relief activities (Sinha et al. 2019). In this regard, wireless sensor networks (WSNs) and unmanned aerial vehicles (UAVs) with aim of collecting real-time information from various sources seem essential. WSNs are networks of sensors that monitor

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and record the physical conditions of the environment to send to the data center. These tools can measure environmental conditions such as noise, temperature, pollution levels, and wind (Ullo and Sinha 2020). Furthermore, UAVs are aircraft without a human pilot and crew that are remotely controlled by a human operator or fully autonomous (Hu et al. 2022). Obviously, the existence of Internet infrastructure is the first step to obtain the benefits of the mentioned tools. There are several case studies on the use of UAV-assisted data collection in wireless sensor networks for monitoring, some of which include Dan et al. (2020), Dong et al. (2014), Frigerio et al. (2014), Chen et al. (2013), and Tuna et al. (2012). In this regard, Twitter, LinkedIn, and Facebook as widely used social networking tools can be used to collect data from people in the disaster area. After that, the data received from disaster areas should be processed immediately. To this end, a cloud center with parallel computing and a high speed of processing can be applied (Nagendra et al. 2020). All of these issues led to the presentation of a method in which gathering online data and processing them in a timely manner, as an important initial step at the beginning of the relief process, the distribution of relief commodities and evacuation of injured people to be done in a more appropriate way. In fact, by taking an idea of the real situation after the earthquake in western Iran and the importance of the time factor in evacuating the affected people and providing appropriate relief, a three-stage model is designed, modeled, and finally optimally solved. Here, online data obtained from disaster areas is processed as an initial stage using a cloud computing center, and the output of this process is used as input parameters in the next step in the process of solving the mathematical model for the optimal distribution of relief and evacuation of injured people.

The lack of proper and adequate data from a disaster area and also the absence of reliable information regarding the exact location of the involved people in the disaster will cause irreparable loss (Sadidi et al. (2018); the aim of this paper, by considering a cloud center, image processing, and tweets mining, is to apply the accurate and online information about disaster areas in the model to improve the relief distribution and wounded transportation. Consequently, one of the important challenges to be answered in this paper is how to decrease the disaster effects and improve the relief operations using recent developments in technology and also with social media data usage. Since quick and effective relief can increase the survival rate of involved people in disaster areas, presenting a comprehensive approach to accomplish these goals seems necessary. In this regard, here, an attempt is made to develop a three-stage procedure to achieve the intended goals. So in this paper, by considering a cloud center in which data processing is done, a three-stage procedure is developed that aims to increase the number of survivals and decrease the unsatisfied demand. The studied disaster network and data flow are shown in Fig. 1. Note that in this figure, the dashed arrows are data flows, and normal arrows show the flow for the transportation of people and relief items.

In general, the contributions of current research can be listed as follows: (*i*) presenting a three-stage procedure for relief logistics and evacuation, (*ii*) considering a cloud center for analyzing the obtained data from disaster areas at high speed, and (*iii*) proposing a mathematical model and image processing as well as Tweets mining simultaneously.

The remaining of this study would be as follows. In the Section "Literature review," the literature review is presented. Our three-stage model is described in the Section "Problem description and mathematical model". In the Section "Results," we illustrate our three-stage model of an earthquake disaster. Managerial insights are provided in the Section "Managerial insights." In the Section "Conclusion, limitations, and future research," we conclude our study by making a brief explanation of the results and paving the way for future research.

Literature review

In this section, the related research will be reviewed. The studies about logistic planning and evacuation will be reviewed first, and after that, the literature on cloud computing (CC) and Big Data (BD) will be reviewed.

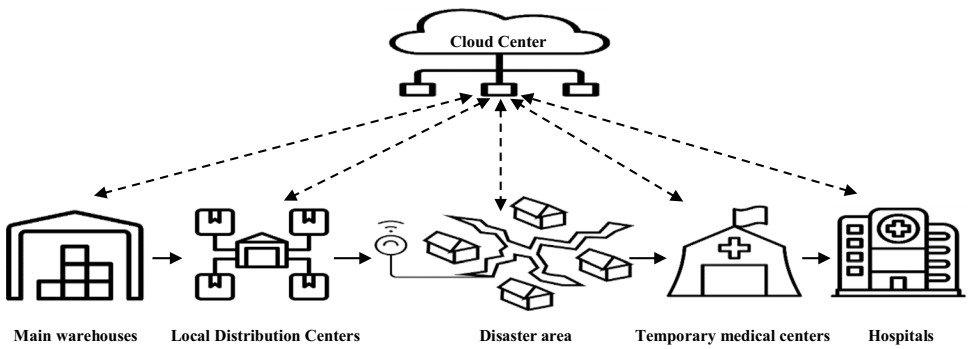


Fig. 1 Disaster networks of the current study

Logistics planning and evacuation

Balcik and Beamon (2008) developed a mixed-integer programming (MIP) model for location and stock pre-positioning of the local distribution centers (LDCs). Their objective was maximizing the expected satisfied demand, and the demand locations and quantity and supply quantity were considered stochastic. Also, Balcik et al. (2008) proposed a multi-period MIP, which combined relief dispatching and resource allocation to determine schedules for available vehicles and a fair resource allocation. The objective of their study was the maximization of the affected people's benefits and the minimization of transportation and unsatisfied demand costs. The problem of emergency vehicle transportation has been solved by Mizumoto et al. (2011) that the objective was to maximize the wounded people's survival by prioritizing the wounded people's transportation. For the multiple-disaster–multiple-response emergency allocation, Zhang et al. (2012) proposed a MIP and heuristic algorithm.

Bozorgi-Amiri et al. (2013) presented a two-stage stochastic model, in which, in the first stage, the needed inventory quantities for each kind of relief commodities and the location of relief distribution centers were determined, and the amount of transportation from distribution centers to disaster areas was determined in the second stage. In order to identify the location of temporary shelters and determine the evacuation routes, Alçada-Almeida et al. (2009) designed a multi-objective location-evacuation model. They considered a limitation for shelter site utilization and a predefined number of temporary shelters. Also, Chanta and Sangsawang (2012) presented a bi-objective model to locate the shelter sites in case of a flood disaster. A mixed-integer linear programming model for the evacuation-location problem was designed by Mazraeh Farahani et al. (2018) to select one or more locations for evacuation. Park et al. (2012) determined the optimal locations of shelters with the objective of minimizing evacuation time. Coutinho-Rodrigues et al. (2012) considered a location-routing problem and identified the shelter locations and evacuation paths by a multi-objective approach. Also, to determine the relief demands and allocate the temporary facilities, Sebatli et al. (2017) presented a simulation-based approach and, also, dispatching the relief commodities stocked in the facilities. The location problem for aerial and ground ambulance bases was considered by Shahriari et al. (2017) as a bi-objective mathematical model for service level maximizing and travel time minimizing. Gharib et al. (2018) designed a three-stage approach for the emergency vehicle routing problem (VRP) based on clustering and considering reliability, and two well-known metaheuristics, i.e., NSGA-II and multi-objective firefly algorithms were developed for solving the model. Sabouhi et al. (2019) developed a mathematical model that integrates the relief distribution and evacuation problems in order to routing and scheduling of vehicles simultaneously for evacuating affected people from disaster areas to shelters and distributing relief commodities among them.

Based on a dynamic network flow model, Shin and Moon (2022) suggested a robust evacuation planning problem to determine the optimal routes for evacuees from a building that has the potential to collapse. Vahdani et al. (2022) studied a humanitarian districted logistics network considering emergency facility location-allocation, redistricting, service sharing, and vehicle routing. They studied robust hybrid optimization for a problem with two types of VRP, land and air routing. Insani et al. (2022) proposed a short-notice flood evacuation model by utilizing public transportation to evacuate an uncertain number of people. In this model, to minimize the number of vehicles required, they split the demands and reused the vehicles.

CC and BD

An important issue when a disaster occurs is collecting and sharing disaster information about affected areas in order to optimal decision-making for relief operations. In this regard, because the infrastructures are destroyed or damaged after a disaster, WSNs and UAVs can be used to receive online data from the affected areas. These results lead to the generating data at high rates through social networks and cell phones of injured people in affected areas as well as WSNs and UAVs, which indicates the necessity of using BD information processing technology. Cloud computing, on the other hand, as a service, has the ability to perform high-speed computing. Due to its independence from the device and location and the fact that the infrastructure may have been destroyed in the affected areas, the need to use CC is quite clear.

The potentials of cloud computing in the improvement of disaster management activities were discussed by Craus and Butincu (2014). They performed data mining techniques and nature-inspired heuristics in cloud services. Sadidi et al. (2018) tried to develop a system for improving the performance of relief operations in damaged areas based on the capabilities of cloud computing. The designed system has two parts: hardware and software.

Purohit et al. (2014) presented machine-learning methods in cloud services to identify affected people's requirements according to social media data. In order to predict earthquake magnitude within the next 7 days, Asencio–Cortés et al. (2018) applied cloud-based BD infrastructure in addition to regression algorithms. Based on CC, Zong et al. (2016) proposed a management platform for natural disasters. In order to develop an early warning flood system, using SMS through IoT, Shalini et al. (2016) used a cloud-based architecture for cooperative flood detection. One of the applications of analyses and processing data is that in the early time of disaster happening, early warning systems

can be developed. According to the received data from social media, the accurate location of the people in the affected areas and, also, the quantity and type of the required relief items can be determined (Truong et al. 2014).

In order to reach reliable prediction results for estimating the earthquake parameters, nearest neighbor searching among a large database of observations can be used even though delay in the processing time related to the precise prediction using a large database can be seen using the earthquake early warning (EEW) systems. In this regard, Yin et al. (2018) suggested using a multidimensional binary search tree to organize large seismic databases. Wang and Qin (2020) specified the unification of cloud-assisted IoT (CIoT) and big data systems for an information recovery system that helps the government in decision-making in disaster situations in a more efficient, faster way. In order to track tweets and for detecting a target event, Sakaki et al. (2010) analyzed the real-time interplay of happenings like earthquakes on Twitter and developed an algorithm. A classifier of tweets according to characteristics like the number of words in a tweet, the keywords, and the content of them was designed to find a target event. Following that, the probabilistic spatiotemporal model was created for the target event, which could find the route and center of the event location. Each Twitter user is recognized as a sensor, and Kalman filtering and particle filtering are used. For flood detection according to IoT, BD, and convolutional deep neural network (CDNN), Anbarasan et al. (2020) suggested some methods and ideas to conquer hardships like collecting, saving, and analyzing BD. Taking the input data from the flood BD is the first step. In the next step, by using a distributed file system that reliably stores large files across machines (HDFS) mapreduce, the repeated data were reduced. Pre-processing of data by using missing value imputation and normalization function is the next step. After that, the focus is on preprocessed data and by applying a combination of attributes method, the rule is created. Finally, the created rules were prepared as the input to the CDNN classifier, which categorizes them as (*i*) chances for the flood happening and (*ii*) no chances for the flood happening. Kumar et al. (2019) concentrated on the application of UAVs with the assistance of an IoT tool to gather the sensed data from the sensor tools in disaster situations. Sinha et al. (2019) stated that as disasters are not predictable, preparedness for reducing the disaster impacts is necessary, and one of the tools for this purpose is IoT-based approaches. So they presented an IoT-based approach to identify the requirements of rescue activities in natural disasters. Nagendra et al. (2020) used a cloud computing platform to study satellite real-time big data about weather information and geospatial data to enhance the communication between rescue teams for humanitarian relief operations. Sadidi et al. (2018) tried to develop a system for improving the performance of relief operations in damaged areas based on the capabilities of cloud computing.

In Table 1, the characteristics of current research and related studies are summarized. As a review of the literature shows, although there are studies on the use of CC and BD in disasters, none has addressed it with a quantitative approach

Table 1 Characteristics of the literature and this study

Reference	Mathematical model	Facility location	VRP	Evacuation	BD	CC	Image processing
Yi and Özdamar (2007)	✓	✓	✓	✓			
Ishii et al. (2002)							✓
Hakdaoui and Emran (2020)							✓
Chang et al. (2007)	✓	✓	✓				
Mizumoto et al. (2011)	✓		✓	✓			
Fiedrich et al. (2000)	✓	✓					
Abdalla and Esmail (2019)					✓	✓	
Huang et al. (2017)						✓	
Pillai et al. (2021)					✓	✓	
Nagendra et al. (2020)					✓	✓	
Sinha et al. (2019)						✓	
Wang and Qin (2020)					✓	✓	
Kim et al. (2019)	✓	✓					
Zhang et al. (2022)	✓		✓				
Voigt et al. (2007)							✓
Zou et al. (2020)						✓	
Gupta et al. (2021)							✓
Current study	✓	✓	✓	✓	✓	✓	✓

and modeling. Obviously, due to the importance of quantitative criteria such as the number of survivals, it is necessary to examine how the use of CC and BD due to cost can be effective in providing services to victims and also reducing the number of survivals along with reducing unsatisfied demand. The CC is independent of location and device, and this ability can improve the relief operations where the planning of relief operations can be done in some locations far away from the disaster areas where the risk of infrastructure disruption is high. Also, the high speed of processing is another advantage of CC, which time is the most important element in a disaster where the number of survivors can be dramatically increased by reducing the time of relief operations. As CC is independent of the device, anywhere and anytime can be accessed, and by the mobility of this technology, the involved staff in relief operations can access the system with just an internet connection. Therefore, the main advantages of using the concept of CC in disaster operations are (*i*) mobility, (*ii*) high speed, (*iii*) easy backup, (*iv*) collaboration, and (*v*) device and location independence.

To the best of our knowledge, there is no study that considers the data analysis to be inputted into the optimization models for relief activities. In other words, analyzing the disaster area by tools such as aerial images and tweets mining in order to produce accurate information about disaster areas and inputting these parameters into the mathematical models for relief distribution and wounded transportation is not studied yet. So in this paper, a research study dealing with relief activities that accounts for analyzing the online data about disaster areas is presented. Note that in all papers in this field, no three-stage model can be seen in which online data are used for improving relief and evacuation operations. In addition, all of them used only cloud computing alone (Ishii et al. 2002) or only image processing (Nagendra et al. 2020).

According to these descriptions, the contributions of this study compared to the current literature can be summarized as follows:

- investigating the relief logistics with CC, BD, and image processing simultaneously;
- proposing a three-stage mathematical model in disaster situations;
- considering the system dynamics as a logistics coordination model which can prioritize the damaged areas by image processing and make dispatch orders for vehicles;
- studying real-world issues in the problem under consideration includes the capacity of commodity type at supply centers, the capacity of medical centers with two cases for transferring the wounded people, the number of wounded people serving in medical centers, queue length in medical centers, waiting time for wounded people at a temporary medical center and each variable and parameters which are related to the wounded people; and
- examining the proposed model using real-world data.

Problem description and mathematical model

Natural disasters have killed 4.2 million people and affected almost 7 billion people, and damage of 3400 billion dollars has been reported around the world (EM-DAT 2020). Furthermore, man-made disasters caused thousands of death of affected people (Hoyos et al. 2015). It is predicted that natural and man-made disasters will increase fivefold in the next 50 years (Thomas and Kopczak 2005). The increasing number of natural disasters and the number of affected people show the necessity for improvement in disaster management activities, and obviously, considering the problems related to the relief operations is very important (Kundu et al. 2022). Since rapid and effective relief operations can increase the number of survivals and also in all disasters, excessive sending of some relief items to the affected areas, while some other items are needed, can cause many problems and increase the unsatisfied demands; it seems necessary to provide a comprehensive approach that can be used to increase the number of survivals and decrease the unsatisfied demands (Maghsoudi and Moshtari 2021).

In this section, a three-stage model was developed for the logistics and evacuation in which, first, the job scheduling for data processing is presented in the first stage. Then, two mathematical models for commodity dispatching and transferring of wounded people are formulated in the second stage. Finally, in the third stage, detailed vehicle loading/unloading schedules are obtained by using a system of equations.

In Fig. 2, the relation between the stages is shown. Note that, in the first stage, data received from disaster areas is continuously in progress. In fact, in this stage, the value of parameters such as the number of people in disaster areas and relief requirements are determined, but these values are updated through the time horizon by receiving the new data from the disaster area. Also, in the second stage, the value of variables such as the number of transferred people and quantity of relief transported to the disaster area are determined, but these values are optimized in the next stage, where new data are received from disaster areas. Finally, in the third stage, the values of variables related to the vehicle load/unload instructions are updated.

Note that the problem under study in this research is based on a real-world case study, and since its optimal solution could be achieved by solving the model, the use of other algorithms was not a good choice, and here, only the GAMS software was used to find the optimal solution of our case study.

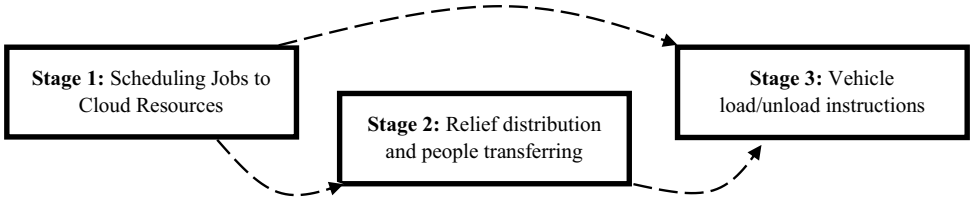


Fig. 2 The relation between the stages in the modeling procedure

Stage I: scheduling jobs on the cloud resources

The jobs which we consider in disasters are multi-stage jobs which means that each job contains several stages to be done. Each stage is a task. Tasks are performed consecutively so that every step contains only one task. Tasks are performed on several parallel processors. In a disaster, the arrival time of multi-stage tasks is considered stochastic. Besides, the processing time of distinct tasks is considered stochastic too. Since it is crucial to perform every task at its predetermined time and delay will result in irreparable damage, a deadline is defined for every task. Zhu et al. (2018) provided a mathematical model and two algorithms for this problem in which we employ their model in this paper. Note that the output of this stage is the number of wounded people and commodity requirements in the disaster area. In other words, image processing and data processing are done in the cloud center, and needed information (number of affected people, commodity requirements, etc.) will be acquired and inserted into the next stage as parameters. Hence, the mathematical model of Zhu et al. (2018) is presented here. Job J_j ($j = 1 \ldots n$) involves m tasks $T_{j,1}$, $T_{j,2}$, …, $T_{j,m}$ which are provisioned by m virtual clusters. One processor is assigned to process each task. Note that in this first stage, all data should be filtered before being collected and processed in the cloud center.

For this purpose, the symbols used, including notation, parameters, and decision variables, are defined as follows.

n	Number of jobs; $j, j' = 1, .., n$
m	Number of stages in a job, $i = 1, \ldots, m$
$b_{j,i}$	The planning horizon length (e.g., a working shift), t = 1, 2, …, T
J_j	j^{th} job
$T_{j,i}$	i^{th} task of J_j
$p_{j,i}$	Processing time of task $T_{j,i}$
t_j	Arrival time of J_j
$b_{j,i}$	Start time of $T_{j,i}$
$c_{j,i}$	Completion time of $T_{j,i}$
$d_{j,i}$	Transferring delay for the intermediate result of $T_{j,i}$
D_j	Deadline of J_j
$v_{j,i}$	Processor assigned to $T_{j,i}$

The mathematical model for processor scheduling with the aim of minimizing the total completion time is as follows.

$$\text{Min} \sum_j \sum_i c_{j,i} \tag{1}$$

$$s.t. : max\{b_{j,i}, b_{j',i}\} \geq min\{c_{j,i}, c_{j',i}\}, v_{j,i} = v_{j',i}, j \neq j' \tag{2}$$

$$c_{j,i} = b_{j,i} + p_{j,i} \tag{3}$$

$$c_{j,i-1} + d_{j,i-1} \leq b_{j,i} \tag{4}$$

$$t_j \leq b_{j,i} \tag{5}$$

$$c_{j,m} + d_{j,m} \leq D_j \tag{6}$$

$$\forall < T_{j,i}, b_{j,i}, v_{j,i} > \in S, j, j' = 1, \ldots, n, i = 1, \ldots, m \tag{7}$$

The objective function of the model in Eq. (1) minimizes the total completion time of all jobs. Constraint (2) is to ensure that on the same processor, no overlapping time occurs among tasks. Constraint (3) implies that preemption is not allowed. The precedence constraint is shown in constraint (4), i.e., until a task is not completed, the next task cannot be started. Constraint (5) implies that a job will not start before it arrives. The deadline constraint is defined in constraint. $> T_{j,i}$, $b_{j,i}$, $v_{j,i} >$ indicates the assignment of the $T_{j,i}$ to processors v_i^k with start time $b_{j,i}$, where $v_{j,i} = v_i^k$. The assigned processor to $T_{j,i}$ is shown by $v_{j,i}$ and a set of assignments $S = \left\{ < T_{j,i}, b_{j,i}, v_{j,i} > | j = 1, \ldots, n, i = 1, \ldots, m \right\}$ which is feasible if and only if the above constraints are satisfied. The complexity of this model is $O(log|S|)$ (Zhu et al. 2018).

Note that, in this paper, it is assumed that there is a cloud center that is able to quickly analyze big data about disaster areas. In fact, the first stage of the three-stage procedure is the scheduling of the parallel processors in the cloud center in which image processing and tweets mining are considered in it. It is expected the cloud center will produce helpful information from online data obtained from disaster areas in a reasonable time which is necessary to improve relief operations. In this stage, the data from disaster areas are

received, which are tweets and images. Each processor has a duty to process these data in order to produce information. Therefore, each processor performs a job and has an output that serves as an input for the next stage. So the parallel processors should be optimally scheduled, and this scheduling is considered as the input to the next steps. Note that the outputs of the first stage are continuously updated and inserted into the next stage. The jobs which are allocated to the processors are tweet mining and image processing, and the output of this stage is information produced from these data. So notice that the scheduling of processors is not the output of the first stage and the scheduling is done in order to speed up the information production. Note that by applying the tweet mining and image processing in this stage, the parameters in the second stage, such as the number of involved people in the disaster areas, their relief needs, the capacity of LDCs and medical centers, etc., are obtained and inputted to the second stage.

Stage II: relief distribution and people transferring

The proposed model aims to interrelate the commodities transportation from supply centers to local distribution centers near the disaster areas and the wounded people's transportation from disaster areas to the permanent (hospitals) and temporary emergency units. The problem notation and mathematical formulation are given below. The most similar work to the present study is Yi and Özdamar (2007) that in a current study several real features have been added to it, including considering the capacity of commodity type at supply centers, the capacity of medical centers with two cases for transferring the wounded people, the number of wounded people serving in medical centers, queue length in medical centers, waiting time for wounded people at a temporary medical center and each variable and parameters which are related to the wounded people.

Notations and parameters

T	The planning horizon length (e.g., a working shift), t = 1, 2, …, T
W	Injured people type
N	Set of nodes
q	Set of supply nodes, q ⊂ N
i	Set of potential temporary dispatching center sites that can be initiated to receive and dispatch the commodities among people in affected regions, i ⊂ N
j	Disaster area (demand node), j ⊂ N
E	Types of demand: E = {1, 2,3}
C	Set of commodities
V	Set of various transportation modes or vehicle types
B	A big number
CV	Vehicles capacity
k	Set of potential temporary medical centers (on-site clinic) which can be initiated in affected regions in order to receive light-moderate injured people, k ⊂ N
m	Set of available permanent medical centers (hospitals) that can receive both light-moderate and heavy types of wounded people, m ⊂ N
SO_{cq}	Capacity of commodity type c at supply center q
D	Case of transporting wounded people straightly from affected regions to hospitals
ID	Case of transporting wounded people from affected regions to the permanent medical center through a temporary medical center
CC_k	temporary medical center k capacity
CH_m	General hospital m capacity
P^t_{wj}	Number of wounded people of type w who are found and saved at affected region j through period t
D^t_{wj}	Number of waiting wounded people of type w at affected region j at period t
TC_w	Waiting time for type w wounded people at a temporary medical center to get first-aid
TO_w	The length of time that wounded people w receive treatment in a general hospital
TM_{jmv}	Time of transportation from affected region j to hospital m by vehicle v
TM_{jkv}	Time of transportation from affected region j to temporary medical center k by vehicle v
TM_{kmv}	Time of transportation from temporary medical center k to permanent medical center m by vehicle v
$f_D(w,t)$	Probability of survival of type w wounded people at period t in case D ($0 \le f_D(w,t) \le 1$)
$f_{ID}(w,t)$	Probability of survival of type w wounded people at period t in case ID ($0 \le f_{ID}(w,t) \le 1$)
w_c	Commodity c unit weight
w_w	Wounded people's average weight
cap_v	Vehicle type v load capacity
f_E	Standard amount of commodity E required to sustain one affected people
P_w	Serving wounded people w priority, which is equal to $P_w = k_w / \sum_t \sum_j D^t_{wj}$ where k_w is a positive subjective parameter

Decision variables

IDS^t_{wj}	Number of wounded people w at affected regions j at the end of period t
IC^t_{wk}	Number of wounded people w at temporary medical center k at the end of period t

ICC^{t}_{wk} Number of wounded people w in the treatment process at temporary medical center k at the end of period t

IH^{t}_{wm} Number of wounded people w at permanent medical center m at the end of period t

IM^{t}_{wjk} Number of wounded people w in the transporting process from affected region j to temporary medical center k at the end of period t

IM^{t}_{wjm} Number of wounded people w in the transporting process from affected region j to permanent medical center m at the end of period t

IM^{t}_{wkm} Number of wounded people w in the transporting process from temporary medical center k to permanent medical center m at the end of period t

WC^{t}_{wk} Waiting number of wounded people w to get first-aid at temporary medical center k at the end of period t

$WH^{t}_{D,wm}$ Waiting number of wounded people w to get treatment at permanent medical center m at the end of period t in case D

$WH^{t}_{ID,wm}$ Waiting number of wounded people w to get treatment at permanent medical center m at the end of period t in case ID

IOH^{t}_{wm} Number of wounded people w in the treatment process at permanent medical center m at the end of period t

$POH^{t}_{D,wm}$ Number of wounded people w who begin to be treated on at permanent medical center m at the end of period t in case D

$POH^{t}_{ID,wm}$ Number of wounded people w who begin to be treated on at permanent medical center m at the end of period t in case ID

PCC^{t}_{wk} Number of wounded people w who begin to receive treatment at temporary medical center k at the end of period t

x^{t}_{wjm} Number of wounded people w moved from affected area j to permanent medical center m through period t

x^{t}_{wjk} Number of wounded people w moved from affected region j to temporary medical center k through period t

x^{t}_{wkm} Number of wounded people w moved from temporary medical center k to permanent medical center m through period t

Q^{t}_{cqi} Commodity c amount conveyed from supply node q to dispatching site i at period t

A^{t}_{cij} Commodity c amount conveyed from dispatching site i to affected region j at period t

Z_i Equal to 1 if distribution center i is selected to establish and 0 otherwise

V^{t}_{qiv} Type v vehicles integer number traversing from supply node q to dispatching site i at period t

V^{t}_{ijv} Type v vehicles integer number traversing from dispatching site i to affected region j at period t

V^{t}_{jkv} Type v vehicles integer number traversing from affected region j to temporary medical center k at period t

V^{t}_{jmv} Type v vehicles integer number traversing from affected region j to permanent medical center m at period t

V^{t}_{kmv} Type v vehicles integer number traversing from temporary medical center k to permanent medical center m at period t

DEV_{Eot} Amount of unsatisfied demand of commodity type E at node o at time t

DEV_{wot} Number of unserved wounded people of category w at node o at time t

The model for transferring the wounded people to temporary and permanent medical centers is presented here, which aims to maximize the number of survivors.

$$maximize \frac{\sum_{t=1}^{T}\sum_{w=1}^{W}\sum_{m=1}^{M} P_w\left(POH^{t}_{D,wm} f_D(\mathrm{w,t}) + POH^{t}_{ID,wm} f_{ID}(w,t)\right)}{\sum_{t=1}^{T}\sum_{w=1}^{W}\sum_{m=1}^{M} P^{t}_{wj}} \tag{8}$$

$$s.t.: IDS^{t}_{wj} = IDS^{t-1}_{wj} + P^{t}_{wj} - \left(\sum_{m} x^{t}_{wjm} + \sum_{k} x^{t}_{wjk}\right) \ \forall w,j,t \tag{9}$$

$$IC^{t}_{wk} = IC^{t-1}_{wk} + \sum_{j} x^{t-TM_{jk}}_{wjk} - \sum_{m} x^{t}_{wkm} \ \forall w,k,t \tag{10}$$

$$IH^{t}_{wm} = IH^{t-1}_{wm} + \sum_{j}\sum_{k} x^{t-TM_{jm}}_{wjm} - \sum_{k} x^{t-TM_{km}}_{wkm} \left(POH^{t-\mathrm{TO}_w}_{D,wm} + POH^{t-\mathrm{TO}_w}_{ID,wm}\right) \ \forall w,m,t \tag{11}$$

$$IM^{t}_{wjm} = IM^{t-1}_{wjm} + x^{t}_{wjm} - x^{t-TM_{jm}}_{wjm} \quad \forall w, j, m, t \tag{12}$$

$$IM^{t}_{wjk} = IM^{t-1}_{wjk} + x^{t}_{wjk} - x^{t-TM_{jk}}_{wjk} \quad \forall w, j, k, t \tag{13}$$

$$IM^{t}_{wkm} = IM^{t-1}_{wkm} + x^{t}_{wkm} - x^{t-TM_{km}}_{wkm} \quad \forall w, k, m, t \tag{14}$$

$$WC^{t}_{wk} = WC^{t-1}_{wk} + \sum_{j} x^{t-TM_{jk}}_{wjk} - PCC^{t}_{wk} \quad \forall w, k, t \tag{15}$$

$$WH^{t}_{D,wm} = WH^{t-1}_{D,wm} + \sum_{j} x^{t-TM_{jm}}_{wjm} - POH^{t}_{D,wm} \tag{16}$$

$$WH^{t}_{ID,wm} = WH^{t-1}_{ID,wm} + \sum_{k} x^{t-TM_{km}}_{wkm} - POH^{t}_{ID,wm} \quad \forall w, m, t \tag{17}$$

$$PCC^{t}_{wk} \le WC^{t-1}_{wk} + \sum_{j} x^{t-TM_{jk}}_{wjk} \quad \forall w, k, t \tag{18}$$

$$POH^{t}_{D,wm} \le WH^{t-1}_{D,wm} + \sum_{j} x^{t-TM_{jm}}_{wjm} \quad \forall w, m, t \tag{19}$$

$$POH^{t}_{ID,wm} \le WH^{t-1}_{ID,wm} + \sum_{k} x^{t-TM_{km}}_{wkm} \quad \forall w, m, t \tag{20}$$

$$ICC^{t}_{wk} = ICC^{t-1}_{wk} + PCC^{t}_{wk} - PCC^{t-TC_{w}}_{wk} \quad \forall w, k, t \tag{21}$$

$$IOH^{t}_{wm} = IOH^{t-1}_{wm} + \left(POH^{t}_{D,wm} + POH^{t}_{ID,wm}\right) - \left(POH^{t-\mathrm{TO}_{w}}_{D,wm} + POH^{t-\mathrm{TO}_{w}}_{ID,wm}\right) \quad \forall w, m, t \tag{22}$$

$$\sum_{w} IC^{t}_{wk} \le CC_{k} \quad \forall t, k \tag{23}$$

$$\sum_{w} IH^{t}_{wm} \le CH_{m} \quad \forall t, m \tag{24}$$

Equation (8) maximizes the number of survivors among wounded people saved in the affected region, which is the objective of the model. Equation (9) shows the number of people waiting for transport from the affected area, and Eqs. (10) and (11) show the number of people waiting for medical services at temporary and permanent medical centers, respectively. The balance equations for changes in the number of wounded people transported from disaster regions to permanent medical centers, from disaster regions to temporary medical centers, and from temporary medical centers to permanent medical centers are described in Eqs. (12) to (14), respectively. The number of wounded people waiting for treatment in temporary and permanent medical centers is explained by Eqs. (15) and (16), respectively. The balance equation for changes in the number of wounded people waiting for treatment in hospitals is shown in Eq. (17). The number of wounded people who can be served for medical services at temporary medical centers is indicated in Eq. (18). Limitations on the number of available wounded people who can receive treatment in cases *D* and *ID* are shown in Eqs. (19) and (20). Equation (21) guarantees the balance equation for changes in the number of wounded people who are receiving treatment. The balance equation for changes in the number of wounded people who are receiving treatment in general hospitals is explained in Eq. (22). Equations (23) and (24) show the capacities of the temporary and permanent medical centers, respectively. The complexity of this model is O(*m*) which *m* indicates the number of limitations.

The relief distribution model.

In this model, the number of commodities to be delivered to affected areas will be determined, and it aims to minimize the unsatisfied demand.

$$min \sum_{E} \sum_{o \epsilon j} \sum_{t} P_{E} DEV_{Eot} \tag{25}$$

$$s.t.: \quad x^{t}_{wjm} + x^{t}_{wjk} + x^{t}_{wkm} \le CV \quad \forall w, j, k, m, t \tag{26}$$

$$\sum_{q} Q^{t}_{cqi} - \sum_{E} \sum_{o \in j} \sum_{t} DEV_{Eot} = \sum_{j} A^{t}_{cij} \quad \forall i, c, t \tag{27}$$

$$\sum_{i} Q^{t}_{cqi} \le SO_{cq} \quad \forall c, q, t \tag{28}$$

$$\sum_{q} Q^{t}_{cqi} \le Cap_{i} \quad \forall c, i, t \tag{29}$$

$$Q^{t}_{cqi} \le B.Z_{i} \quad \forall c, q, i, t \tag{30}$$

$$A^{t}_{cij} \le B.Z_{i} \quad \forall c, q, i, t \tag{31}$$

$$V^{t}_{jkv} \le MTM_{jk} \quad \forall j, k, v \tag{32}$$

$$V^{t}_{jmv} \le MTM_{jm} \quad \forall j, m, v \tag{33}$$

$$V^{t}_{kmv} \le MTM_{km} \quad \forall k, m, v \tag{34}$$

$$\sum_{v} V^{t}_{qiv} cap_{v} \ge \sum_{c} w_{c} Q^{t}_{cqi} \quad \forall q, i, t \tag{35}$$

$$\sum_{v} V^{t}_{ijv} cap_{v} \ge \sum_{c} w_{c} A^{t}_{cij} \quad \forall i, j, t \tag{36}$$

$$\sum_{v} V^{t}_{jkv} cap_{v} \ge \sum_{w} w_{w} x^{t}_{wjk} \quad \forall j, k, t \tag{37}$$

$$\sum_{v} V^{t}_{jmv} cap_{v} \geq \sum_{w} w_{w} x^{t}_{wjm} \quad \forall j, m, t \tag{38}$$

$$\sum_{v} V^{t}_{kmv} cap_{v} \geq \sum_{w} w_{w} x^{t}_{wkm} \quad \forall k, m, t \tag{39}$$

Equation (25) minimizes the weighted sum of unsatisfied demand. Equation (26) states that the entire number of affected people who can be transported in period t may not be more than the available vehicles' total capacity. Equation (27) states that all commodities which are sent to an LDC must leave that center. Equation (28) shows that relief commodities that are sent from a supply node to LDCs cannot exceed the capacity of the supply node. Equation (29) shows the capacity limitation of relief distribution centers. Equations (30) and (31) state that if there is a distribution center i then transferring the commodities is possible. Constraints (32–34) restrict the itinerary of each vehicle type for transferring wounded people to existing paths. Constraints (35–39) determine the quantities of transportation by the vehicles' capacity.

Stage III: vehicle load/unload instructions

In this stage, by solving a set of linear equations, the loaded/unloaded quantities of vehicles are calculated. This system of equations is easier to solve compared to the model of the second stage (Yi and Özdamar 2007); however, it includes integer variables for wounded people. Matrix S is defined in which element $S[v,o,t]$ shows that a vehicle v is at a node o in time t. Since a node can be met more than once by vehicle v during the planning horizon, the period t must be defined. In the previous stage, the quantity of transported relief items and also the number of transferred people are determined and this shows the location of vehicles in the network. In this regard, because the information of disaster is updated continuously through the time horizon, it is possible that the route for every vehicle changes. For example, the first route for vehicle 1 is to move through nodes 1–2-3, and when the vehicle arrives at node 2, according to the new information, the capacity of node 3 is completed. The new dispatch order receives, and the vehicle should go to node 4. So this matrix is an output of stage 2 which shows the location of each vehicle at each time in the network.

In the following, the system of equations is provided, which recognizes commodities quantity and the number of wounded people loaded and unloaded by each vehicle at every node. The same parameters as the second stage are used, and new decision variables are defined as follows.

Decision variables

QP^{t}_{cov}
Commodity c amount picked up in period t by vehicle v at node o ($o \subset q, i$)

QD^{t}_{cov} Commodity c amount delivered in period t by vehicle v at node o ($o \subset i, j$)

xp^{t}_{wov} Number of wounded people w picked up in period t by vehicle v at node o ($o \subset j, k$)

Xd^{t}_{wov} Number of wounded people w delivered in period t by vehicle v at node o ($o \subset k, m$)

System of equations

$$\sum_{v} \sum_{t=1}^{T-1} \left[xp^{t}_{wov} - xd^{t}_{wov} \right] = \sum_{j} \sum_{m} \sum_{t=1}^{T-1} \left[x^{t}_{wjk} - x^{t}_{wkm} \right] \quad \forall o, w, k \tag{40}$$

$$\sum_{v} \sum_{t=1}^{T-1} \left[QP^{t}_{cov} - QD^{t}_{cov} \right] = \sum_{q} \sum_{j} \sum_{t=1}^{T-1} \left[Q^{t}_{cqi} - A^{t}_{cij} \right] \quad \forall c, o, i \tag{41}$$

$$\sum_{o} \sum_{t=1}^{T} \left[xp^{t}_{wov} - xd^{t}_{wov} \right] = 0 \quad \forall v, w \tag{42}$$

$$\sum_{o} \sum_{t=1}^{T} \left[QP^{t}_{cov} - QD^{t}_{cov} \right] = 0 \quad \forall c, v, i \tag{43}$$

$$\sum_{w} \left[xp^{t}_{wov} + xd^{t}_{wov} \right] + \sum_{c} QP^{t}_{cov} + QD^{t}_{cov} \leq S[v, o, t].B \quad \forall o, v, t \tag{44}$$

$$\sum_{w} \sum_{o} \sum_{t=1}^{T} W_{w} \left[xp^{t}_{wov} - xd^{t}_{wov} \right] + \sum_{c} \sum_{o} \sum_{t=1}^{T} W_{c} \left[QP^{t}_{cov} - QD^{t}_{cov} \right] \leq cap_{v} \quad \forall v \tag{45}$$

$$xp^{t}_{wov}, xd^{t}_{wov} \geq 0 \text{ and integer}; QP^{t}_{cov}, QD^{t}_{cov} \geq 0 \tag{46}$$

Constraint (40) indicates the optimal net picked-up wounded people quantity found in the second stage must be equal to the net cumulative number of wounded people type w picked up from node o by all vehicles in each period. Constraint (41) is the same as constraint (40) and applies to commodities. Constraints (42) and (43) guarantee that the picked-up quantities by each vehicle

are equal to the delivered quantities at the end of the planning horizon. If and only if a vehicle is at a node in period t, it can pick up and deliver from/to that node, and this is shown in constraints (44). Constraint (45), as a capacity constraint, shows the net aggregative picked-up quantity by a vehicle at any time through all nodes is limited by its capacity. The variable domains are defined in constraint (46). By solving the above set of equations, the optimal picked-up and delivered quantities by each vehicle are recognized. The complexity of this model is $O(V|N^2|T)$, in which V is the total number of vehicles; N shows the number of nodes, and it has a pseudo-polynomial complexity.

Results

As shown in Fig. 3, we apply the three-stage model to the disaster of the 2017 Sarpol-e-Zahab earthquakes in Iran. At least 620 people died, and more than 3000 others were injured that needed emergency treatment and were categorized by injury types. Two permanent medical centers (hospitals) and twelve temporary medical centers (on-site clinics) were available, which can be seen in Figs. 3 and 4.

In the first stage, the jobs, as multi-stage tasks, including image processing in order to know the damages in affected areas and processing of the Tweets in order to identify the casualties and relief requirements of affected people. For image processing, the method developed by Ishii et al. (2002) is employed to detect earthquake-damaged areas from aerial photographs. They use color and edge information to detect the damaged area. The proposed two methods are (1) one aerial photo after the disaster and (2) two photos of the same area before and after the disaster.

In Fig. 5 aerial photo, which is one of the aerial photos of the disaster area, the red areas indicate the damaged areas; the purple part represents the hospital, and the green areas show the temporary shelters. Also, a mining tool called Tweedr, which extracts information to help disaster responders, is designed by Ashktorab et al. (2014). Classification, clustering, and extraction are the three main parts of the Tweedr. In the first phase, which is classification, different classification methods are employed to identify keywords in tweets. In the next phase (clustering), in order to cluster similar tweets, some filters are used, and in the last phase (extraction), the phrases and words which show specific information about

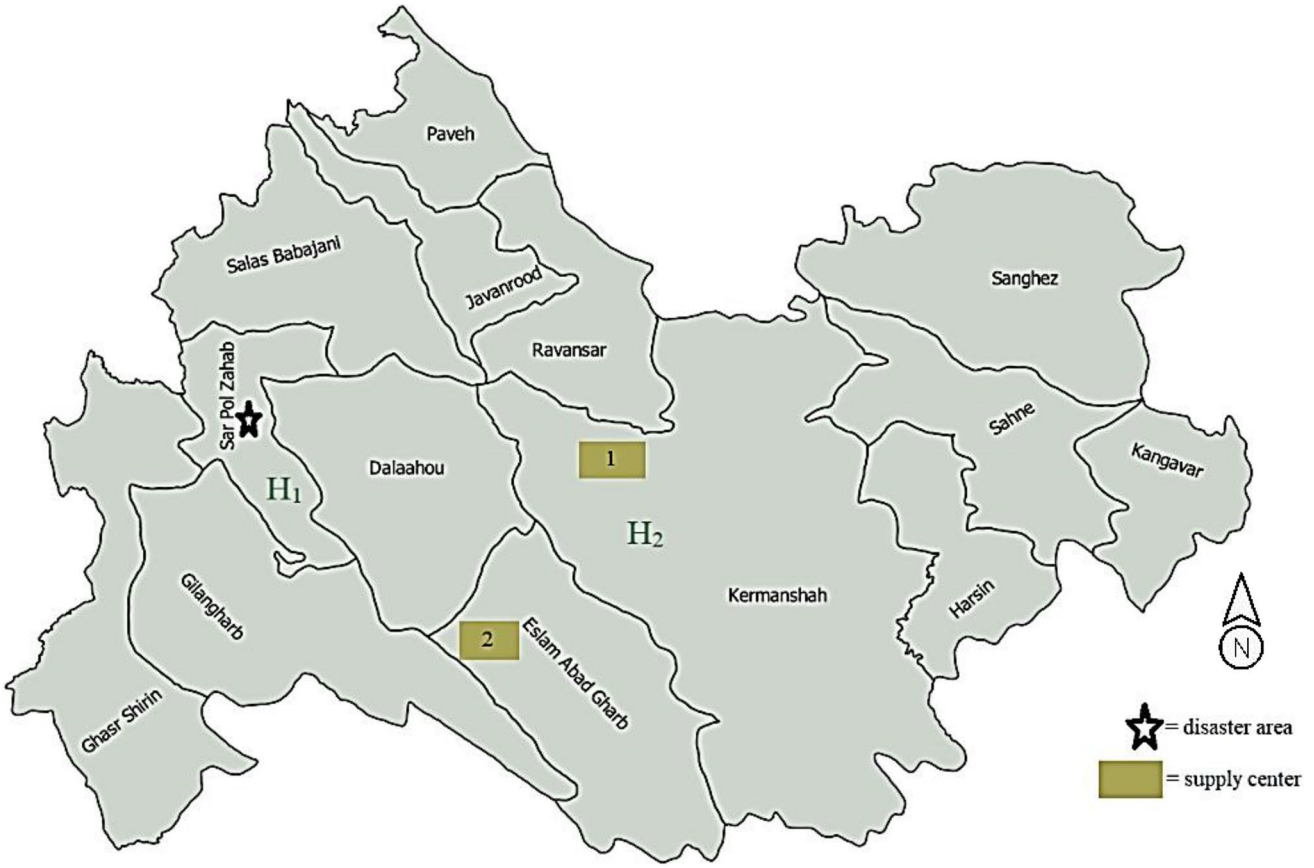


Fig. 3 The location of the disaster area, available hospitals, and supply centers. Note that, in Fig. 4, the location of LDC 9 and 10 is not on the map of Sarpol-e-Zahab

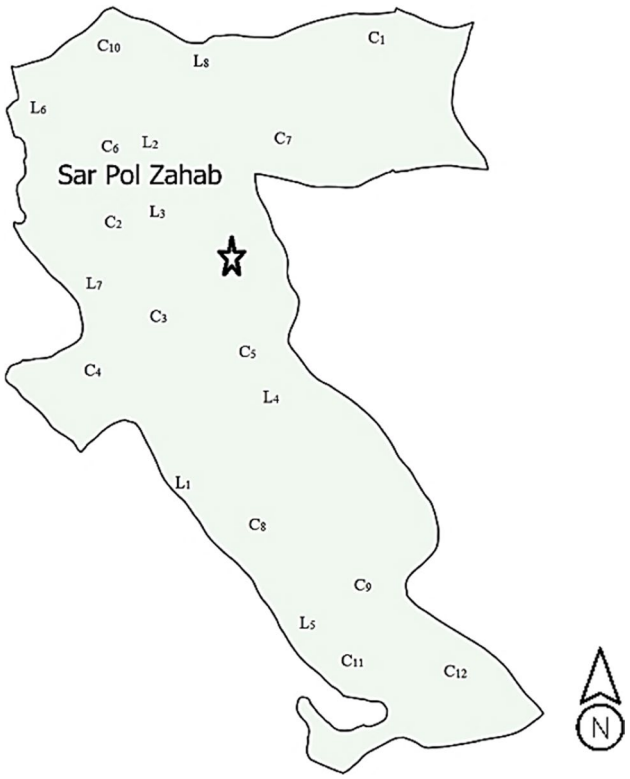


Fig. 4 The location of on-site clinics and LDCs

the disaster are extracted. The Tweedr is available at https://github.com/dssg/tweedr. We consider some keywords in the processing of the tweets in Persian: "earthquake," "we are here," "water," and "bread" are some of these keywords.

Applying these processes in the first stage, the output which is the input for the second stage is summarized in Tables 2, 3, 4, 5, 6, 7, 8, 9, 10, and 11. Note that this information (e.g., Table 2) is generated based on the data received from the disaster area. By image processing and tweet mining as well as initial data received from the disaster area, the number of wounded people will be available. For example, by considering a specific filter in tweets, the number of people who are stuck in a particular area can be obtained. In Table 3, since transferring people to on-site clinics by helicopter is not allowed, the time of transfer is neglected, as same as the time of transportation between temporary and permanent medical centers in Table 4. Note that, as there is a cloud center, both models are solved simultaneously in the cloud center by the high speed of processing.

The maximum capacity at hospitals and on-site clinics is shown in Table 5.

The processing time at each hospital and the on-site clinic is indicated in Table 6.

The distribution of commodity demand and supply is indicated in Tables 7, 8, 9 and 10 respectively.

We consider two major relief commodities (E_1 and E_2) and medicine which, for convenience, their supplies and demands are provided in units of people equivalent. Note that we categorize relief commodities into two major



Fig. 5 Aerial photo of Sarpol-e-Zahab after the earthquake (source of image: Google Earth)

Table 2 Number of wounded type *w* found and saved at affected area *j* in time *t*

		Type of injury			
		W_1	W_2	W_3	W_4
Time	1	92	133	135	162
	2	43	92	86	150
	3	24	80	96	103

groups: type 1 (E1) items whose demand occurs once at the beginning of the planning horizon (e.g., tarpaulins, blankets, tents, jerry cans). Type 2 (E2) are the items that are used regularly and repeatedly happen on the planning horizon (e.g., hygiene kits, food).

In Table 11, the capacity of the vehicle for each vehicle type is shown. In this instance, the total number of vehicles is 3525, and trucks are the most available vehicles.

The results of stage II and stage III are summarized in Tables 12, 13, 14, 15, and 16. All models are implemented on a PC with a CPU of 1.66 GHz, core i5, and 4 GB of memory and are solved with the CPLEX solver in the GAMS software. In Table 12, the utilization of the vehicles during the periods is shown. The resource constraints, along with trucks' additional (excessive) transport capacity, cause the utilization ratio of trucks to be fewer than helicopters and ambulances.

The number of injured people transported from the disaster areas to hospitals and on-site clinics at each time is indicated in Table 13. Note that transporting from affected regions to on-site clinics is done by ambulances, and to general hospitals is done by helicopters. As we can see from Table 13, on-site clinics C_1 and C_8–C_{11} are not utilized, and it is because of the distance of these facilities from the disaster areas. It is necessary to mention if the infrastructures between affected regions and other on-site clinics are disturbed: unused facilities can be used.

The number of waiting people (queue lengths) at hospitals and on-site clinics is indicated in Table 14. Compared to the total number of wounded people carried from the disaster areas (Table 13) to medical facilities, the queue sizes are negligible. In Table 14, a decrease is seen in the queue lengths in the third period since the system is steady.

Table 4 The estimated time of transportation between hospitals and on-site clinics

On-site clinic	*By helicopter*		*By ambulance*	
	General hospital		General hospital	
	H_1	H_2	H_1	H_2
K_1	-	23	15	135
K_2	-	28	17	137
K_3	-	43	10	142
K_4	-	39	12	130
K_5	-	33	19	143
K_6	-	37	12	141
K_7	-	40	20	148
K_8	-	42	24	140
K_9	-	37	30	135
K_{10}	-	34	22	132
K_{11}	-	42	15	139
K_{12}	-	40	18	120

The number of commodities E_2 transferred from supply nodes to LDCs at each time is presented in Table 15.

As indicated in Tables 15 and 16, LDCs 1, 5, 9, and 10 are not selected because of their distance to disaster areas. Also, it can be seen that supply centers transfer the major supplies to the LDCs 3 and 4, and it is because of this fact that these LDCs have a short distance from the disaster areas, and most of the supplies from supply centers 1 and 2 are transferred to LDC 3 because this LDC is near the disaster area.

Managerial insights

A quick response when a disaster occurs can considerably decrease the loss rate. Moreover, having reliable online data from disaster areas can significantly help to reduce the time and improve the quality of relief operations. By analyzing these data, which can be obtained from resources such as WSNs, UAVs, and social networks, people's requirements in disaster areas and also their exact location of them can be accessed. The amount of data produced and gathered

Table 3 Estimated time of transportation (minute) between the affected area and medical centers

	General hospital		On-site clinic											
Vehicle	H_1	H_2	K_1	K_2	K_3	K_4	K_5	K_6	K_7	K_8	K_9	K_{10}	K_{11}	K_{12}
Ambulance	63	120	38	27	14	20	17	21	19	37	40	42	39	14
Helicopter	35	58	-	-	-	-	-	-	-	-	-	-	-	-

Table 5 Maximum capacity at temporary medical centers (people)

	General hospital		On-site clinic						
	H_1	H_2	K_1	K_2	K_3	K_4	K_5	K_6	K_7
Maximum capacity	90	85	18	15	26	20	17	14	22

Table 6 Processing time at temporary and permanent medical centers clinics by injury category (minute)

	Type of injury			
	w_1	w_2	w_3	w_4
On-site clinic	4	15	17	-
General hospital	-	-	-	34

in disasters is huge, and all of this takes place within a short period of time. The sources of these data are Social Media, UAVs, WSNs, etc. One of the applicable tools, in order to analyze big data, is cloud computing which can help the data scientist to make knowledge from raw data in a short time which is important to alleviate the disaster impacts. In this paper, by presenting a three-stage approach, it has been tried to apply online data in order to obtain the amount of people's needs and also by image processing safe locations to establish temporary shelters and as well as damaged areas can accurately be determined. Since there was no three-stage approach in the literature, here, the obtained results of the presented approach were compared with Yi and Özdamar (2007). In this regard, in Table 17, quantities of unsatisfied demand in each period are shown for the three-stage procedure developed in this study and the two-stage approach presented by Yi and Özdamar (2007). As it can be seen from this table, the proposed procedure has significantly better performance. The total number of unsatisfied demand in the last row shows that for all three types of commodities, the proposed approach outperformed Yi and Özdamar (2007). The better performance is due to the fact that online data are continuously received through the planning horizon, and relief items are dispatched more effectively. Note that in Tables 17, and 18, model 1 is used for the developed approach in this study, and model 2 is used for the results of Yi and Özdamar (2007). Also, the number of surviving people is shown in Table 18.

It can be seen from this table that the number of surviving people in the proposed model is significantly more than the other one, and in this regard, what is obvious is that the most important issue in all disasters is to save the lives of affected people. Figure 6 confirms this issue, but it is necessary to mention one point, although cloud computing can offer numerous advantages to different stakeholders, using it also has challenges that should be considered by managers in its application. It is notable that with the support of deployment and service models of CC, challenges can be either minimized or eliminated because the dynamic characteristics of CC can always support users. The amount of data produced and gathered in disasters is huge, and all of this takes place within

Table 7 Distribution of commodity demand (people equivalent)

		Time and Commodity								
		$t=1$			$t=2$			$t=3$		
		E_1	E_2	Medicine	E_1	E_2	Medicine	E_1	E_2	Medicine
Node	Disaster area	450	33,040	1760	335	25,930	1110	273	24,390	240
	K_1	-	1300	320	-	1100	240	-	850	200
	K_2	-	1230	300	-	1120	210	-	930	190
	K_3	-	1150	350	-	1070	220	-	870	200
	K_4	-	1220	250	-	1135	200	-	900	180
	K_5	-	1200	250	-	1125	180	-	1010	150
	K_6	-	570	210	-	490	160	-	410	130
	K_7	-	1130	260	-	1100	110	-	870	90
	K_8	-	900	200	-	790	210	-	1000	120
	K_9	-	1100	180	-	930	230	-	830	145
	K_{10}	-	1230	120	-	1200	100	-	910	230
	K_{11}	-	600	200	-	870	90	-	720	175
	K_{12}	-	810	130	-	540	110	-	650	100
	H_1	-	-	1420	-	-	1380	-	-	1300
	H_2	-	-	1350	-	-	1230	-	-	1275

Table 8 Supply distribution of E_1 (people equivalent)

Time	Supply center		
	1	2	3
1	110	150	130
2	85	75	100
3	93	87	95
Total supply	268	282	295

Table 9 Supply distribution of E_2 (people equivalent)

Time	Supply center		
	1	2	3
1	7500	5800	3300
2	4630	2680	10,020
3	2120	1500	7580
Total supply	14,250	9980	20,900

Table 10 Supply distribution of medicine (people equivalent)

Time	Supply center		
	1	2	3
1	820	675	555
2	610	350	1080
3	295	1250	155
Total supply	1725	2275	1790

Table 11 Fleet composition and the total capacity of the vehicle

Vehicle availability (%)	Total capacity				
	W_2+W_3	W_4	E_1	E_2	Medicine
Helicopters (10.23)	960	605	Rarely	Rarely	Ample
Trucks (77.37)	-	-	Ample	Ample	Rarely
Ambulance (12.4)	1430	530	-	-	-

Table 12 The number of vehicles utilized each time

Time	Helicopters	Ambulances	Trucks
1	106	125	700
2	74	98	750
3	110	115	650

a short period of time. The sources of these data are Social Media, UAVs, WSNs, etc. One of the applicable tools, in order to analyze big data, is cloud computing which can help the data scientist to make knowledge from raw data in a short time which is important to alleviate the disaster impacts.

The results obtained from this research have some limitations, especially in developing countries. For example, there are few official statistics for the use of social media in developing countries (e.g., WhatsApp seems to be a better alternative than Twitter). At the same time, the existence of technologies related to UAVs suitable for disaster conditions, as well as the existence of WSNs and powerful data cloud databases that have the ability to quickly process this volume of data are other limitations of the application of this research in developing countries.

Conclusion, limitations, and future research

Due to the uncertainty of the damaged areas and the number of affected people and the relief requirements after natural disasters, in the first stage, we applied a job scheduling problem in which jobs are image processing and tweets mining. By doing this, some parameters such as the location of damaged areas and the number of affected people as the output of this stage are inputted to the next stage. Then, in the second stage, two mathematical models are proposed for relief commodity and wounded people delivery with capacity constraints. Maximizing the number of survivors among wounded people and minimizing the unsatisfied demand in disaster areas were the objective functions of these models. Also, the models of this stage select the location of possible on-site clinics and LDCs near disaster areas. The output of the second stage is processed to obtain a detailed load instruction sheet in the next stage. Then, in stage III, by using a system of equations, a detailed vehicle loading/unloading schedule was obtained. To validate the proposed three-stage model, it was applied to a case study of the earthquake disaster in Iran. The results showed that the model could select the most appropriate facilities, determine the number of wounded people transported by the vehicles, determine the number of people waiting for a treat at medical facilities, determine the number of commodities transported from supply centers to LDCs, and the number of vehicles utilized at each time. As there is no similar three-stage procedure for the relief operations in the literature, our three-stage approach was compared with a two-stage approach from the literature. The results showed that the proposed method performed better so that, due to accessibility and availability of online information, the total number of unsatisfied demand for all three types of commodities in our approach outperformed the other one.

Table 13 The number of wounded people transported from the disaster area to the medical facilities at each time

Time	K_1	K_2	K_3	K_4	K_5	K_6	K_7	K_8	K_9	K_{10}	K_{11}	K_{12}	H_1	H_2
1	-	189	19	35	44	53	17	-	-	-	-	17	56	70
2	-	114	15	43	23	65	23	-	-	-	-	14	58	65
3	-	108	16	40	37	74	16	-	-	-	-	11	30	50

Table 14 The number of waiting people (queue lengths) at medical facilities at different times

Time	K_1	K_2	K_3	K_4	K_5	K_6	K_7	K_8	K_9	K_{10}	K_{11}	C_{12}	H_1	H_2
1	-	22	10	26	20	31	7	-	-	-	-	17	56	70
2	-	17	7	31	13	37	9	-	-	-	-	14	58	65
3	-	10	5	20	8	25	4	-	-	-	-	11	30	50

Table 15 The number of commodities E_2 transferred from supply node 1 to LDCs at each time

	Time	LDC									
		1	2	3	4	5	6	7	8	9	10
From supply center 1 to	1	-	2500	7500	6000	-	1000	800	700	-	-
	2	-	1300	5700	5200	-	950	800	640	-	-
	3	-	1070	5000	4700	-	500	500	350	-	-

Table 16 The number of commodities E_2 transferred from supply node 2 to LDCs at each time

	Time	LDC									
		1	2	3	4	5	6	7	8	9	10
From supply center 2 to	1	-	1700	6800	5600	-	800	500	400	-	-
	2	-	1300	5100	4050	-	950	700	580	-	-
	3	-	1120	4550	3100	-	970	900	860	-	-

Also, the number of surviving people in the three-stage model was significantly more than in the two-stage model.

This model is applied to an earthquake case study, but as was mentioned, since this model is able to make important decisions, it can also be applied to other disasters optimally. We are interested in expanding our study to the different natural disasters with different scales. However, solving mathematical models has limitations due to the level of complexity of the problem. Here, too, due to the complexity of the considered problem, it is obvious that for natural disasters with a greater extent, we are faced with more complex instances and in order to solve them, a heuristic/metaheuristic needs to be used to find good solutions in a reasonable time. However, the unique feature of solving mathematical models is the optimality of its results, which is always considered by researchers due to limited resources. Another limitation of the problem under consideration in the mathematical model is that transferring commodities between LDCs and transferring injured people between medical centers are not allowed. The next limitation of the problem is that changing the transportation mode is not possible, which means that, for example, for commodity transportation, using a helicopter halfway and then a truck is not allowed. Our suggestions for future studies are to incorporate these limitations into the proposed model and to provide efficient algorithms for the approximate solving of larger natural disasters in a reasonable time.

Table 17 The unsatisfied demand in the proposed model compared to Yi and Özdamar (2007)

		Time and unsatisfied demand								
		t = *1*			*t* = *2*			*t* = *3*		
		E1 (model 1- model 2)	E2 (model 1- model 2)	Medicine (model 1- model 2)	E1 (model 1- model 2)	E2 (model 1- model 2)	Medicine (model 1- model 2)	E1 (model 1- model 2)	E2 (model 1- model 2)	Medicine (model 1- model 2)
Node	Disaster area	113–175	6349–7281	112–175	80–133	5947–6909	78–123	34–72	6242–7214	53–82
	K1	-	643–712	43–76	-	352–397	49–68	-	371–412	31–49
	K2	-	471–542	57–83	-	297–354	37–59	-	274–327	22–43
	K3	-	512–578	61–78	-	392–461	52–86	-	403–472	39–71
	K4	-	781–843	39–54	-	349–403	31–47	-	354–459	20–35
	K5	-	594–647	44–68	-	212–307	24–41	-	237–342	17–29
	K6	-	72–134	81–117	-	178–241	39–72	-	204–265	26–42
	K7	-	482–597	72–103	-	491–579	43–61	-	517–592	31–49
	K8	-	102–186	59–81	-	326–381	33–46	-	327–397	25–36
	K9	-	319–393	42–63	-	431–489	41–54	-	440–502	30–46
	K10	-	467–524	49–68	-	513–581	37–52	-	612–654S	28–43
	K11	-	127–172	62–97	-	417–459	29–43	-	449–512	23–31
	K12	-	376–437	71–117	-	225–267	21–37	-	317–375	17–27
	H1	-	-	68–123	–	-	21–43	-	-	32–49
	H2	-	-	40–89	–	-	27–52	-	-	37–56
Total		113–175	11,180–13,046	900–1392	80–133	10,130–11,828	562–894	34–72	10,720–12,523	431–688

Table 18 The number of surviving people for the developed approach in this study and Yi and Özdamar

The number of surviving people		t_1	t_2	t_3
	Model 1	471	334	285
	Model 2	432	309	263

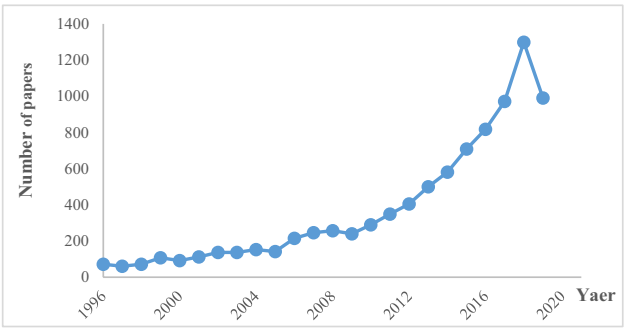


Fig. 6 The number of papers dealing with the application of CC and BD in disasters, source: author's compilation

Author contribution All authors contributed to the study conception and design. Method implementation and optimization, analysis, and data evaluation were performed by MN. The first draft was written by MN, and the final version was edited by JB. All authors read and approved the final submitted manuscript.

Availability of data and materials The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval Not applicable.

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Competing interests The authors declare no competing interests.

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